GATE ME 90 Days Challenge

Day 59/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Manufacturing	**Machining Mechanics & Tool Life**	**43 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on orthogonal and oblique cutting, tool signature (ASA and ORS), chip formation and the built-up edge, shear angle, shear strain and velocity relations, Merchant's circle and force analysis, tool wear and tool materials, Taylor's tool life equation, machinability, cutting power and the economics of machining. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

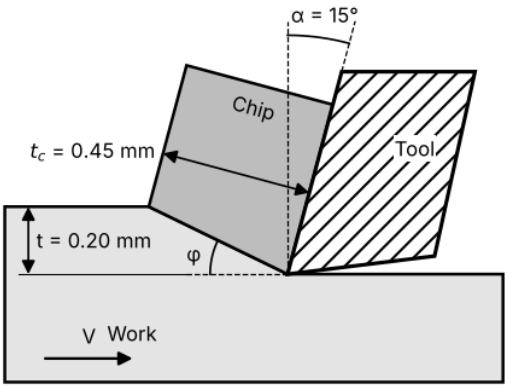
Q1 MCQ

In oblique cutting with a tool of inclination angle i, Stabler's flow rule states that the chip flow angle η (measured on the rake face from the normal to the cutting edge) is

(A) equal to the rake angle
(B) zero, whatever the inclination
(C) approximately equal to the inclination angle i
(D) equal to 90° − i

Q2 NAT

In orthogonal cutting with a tool of rake angle 15°, the uncut chip thickness is 0.20 mm and the chip thickness is 0.45 mm, as shown. The shear angle (in degrees, round off to two decimal places) is _____



Q3 MCQ

In orthogonal cutting with a rake angle of 10°, the shear angle is 30°. The shear strain in the chip is about

(A) 1.73
(B) 2.10
(C) 2.46
(D) 1.36

Q4 MCQ

A single-point tool is specified in the ASA system as 10-12-8-6-15-20-0.8. The end cutting edge angle of the tool is

(A) 10°
(B) 20°
(C) 8°
(D) 15°

Q5 NAT

In orthogonal cutting with a tool of rake angle 8°, the cutting force is 1500 N, the thrust force is 600 N and the chip thickness ratio is 0.32. The shear force along the shear plane (in N, round off to one decimal place) is _____

Q6 MCQ

According to Merchant's theory, if the friction at the tool–chip interface increases while the rake angle remains the same, then

(A) the shear angle decreases, the chip becomes thicker and the cutting force increases
(B) the shear angle increases and the cutting force decreases
(C) the shear angle does not change because it depends only on the rake angle
(D) the chip thickness ratio increases

Q7 NAT

In orthogonal cutting at a cutting speed of 120 m/min with a tool of rake angle 5°, the chip thickness ratio is 0.35. The shear velocity (in m/min, round off to one decimal place) is _____

Q8 NAT

For a tool, the Taylor exponent n is 0.2. The tool life is 30 min at a cutting speed of 100 m/min. The cutting speed (in m/min, round off to two decimal places) that gives a tool life of 60 min is _____

Q9 MCQ

For finish turning of a hardened alloy steel (about 60 HRC), the most suitable cutting tool material is

(A) high-speed steel
(B) cubic boron nitride (CBN)
(C) uncoated cemented carbide of the K (cast iron) grade
(D) polycrystalline diamond (PCD)

Q10 NAT

The Taylor equation for a tool is $VT^{0.25} = 250$ (V in m/min, T in min). The machine and labour cost is ₹4 per minute, the tool cost per cutting edge is ₹40 and the tool changing time is 1 min. The cutting speed for minimum cost per piece (in m/min, round off to one decimal place) is _____

Q11 MCQ

The machinability index (rating) of a material is usually defined as

(A) the hardness of the material relative to that of a free-cutting steel
(B) the ratio of cutting force on the material to that on a free-cutting steel
(C) the surface roughness obtained relative to a free-cutting steel
(D) the cutting speed for a fixed tool life (often 60 min) on the material, as a percentage of that for a standard free-cutting steel

---

Q12 NAT

A bar of 60 mm diameter is turned at 500 rpm with a feed of 0.2 mm/rev and a depth of cut of 1.5 mm. If the specific cutting energy of the work material is 3 J/mm³, the cutting power (in kW, round off to two decimal places) is _____

Q13 MCQ

In machining a ductile metal at moderate to high cutting speed, most of the heat generated is

(A) conducted into the workpiece

(B) carried away by the chip (typically about 70–80 %)

(C) conducted into the tool

(D) radiated to the surroundings

Q14 NAT

A tool obeys the extended Taylor equation $VT^{0.25}f^{0.6} = C$. If the feed is halved, the percentage increase in cutting speed that keeps the same tool life (round off to one decimal place) is _____

Q15 MCQ

In orthogonal cutting, if the rake angle of the tool is increased while the other conditions remain the same,

(A) both the chip thickness ratio and the shear angle increase, and the cutting force decreases

(B) both the chip thickness ratio and the shear angle decrease

(C) the shear angle increases but the chip thickness ratio decreases

(D) the chip thickness ratio increases but the cutting force also increases

Answer Key

Q	Type	Answer
Q1	MCQ	(C) approximately equal to the inclination angle i
Q2	NAT	25.88 (accept 25.70 to 26.06)
Q3	MCQ	(B) 2.10
Q4	MCQ	(D) 15°
Q5	NAT	1234.9 (accept 1228.7 to 1241.0)
Q6	MCQ	(A) the shear angle decreases, the chip becomes thicker and the cutting force increases
Q7	NAT	123.6 (accept 122.4 to 124.9)
Q8	NAT	87.06 (accept 86.62 to 87.49)
Q9	MCQ	(B) cubic boron nitride (CBN)
Q10	NAT	104.3 (accept 103.3 to 105.3)
Q11	MCQ	(D) the cutting speed for a fixed tool life (often 60 min) on the material, as a percentage of that for a standard free-cutting steel
Q12	NAT	1.41 (accept 1.40 to 1.43)
Q13	MCQ	(B) carried away by the chip (typically about 70–80 %)
Q14	NAT	51.6 (accept 51.1 to 52.1)
Q15	MCQ	(A) both the chip thickness ratio and the shear angle increase, and the cutting force decreases

Detailed Solutions

Q1 **Answer: (C) approximately equal to the inclination angle i**

Concept: In oblique cutting the cutting edge is inclined at angle i to the normal of the cutting velocity, so the chip does not flow straight up the rake face.

Stabler found experimentally that the chip flows sideways at an angle $\eta \approx i$, measured on the rake face from the normal to the edge: **(C)**.

In orthogonal cutting $i = 0$, so $\eta = 0$ and the chip flows normal to the edge.

Exam tip: Oblique cutting has three force components (cutting, feed and radial); orthogonal cutting has only two.

Q2 **Answer: 25.88 (accept 25.70 to 26.06)**

Concept: Volume constancy along the shear plane gives $t = l \sin\varphi$ and $t_c = l \cos(\varphi - \alpha)$, so $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$, with $r = t/t_c$.

$r = 0.20/0.45 = 0.4444$

$\tan\varphi = 0.4444 \times \cos 15°/(1 - 0.4444 \times \sin 15°) = 0.4293/0.8850 = 0.4851$

$\varphi = \tan^{-1} 0.4851 =$ **25.88°**

Exam tip: r is always less than 1 (the chip is thicker than the uncut layer), and a larger r means a larger shear angle and lower forces.

Q3 **Answer: (B) 2.10**

Concept: Shear strain in orthogonal cutting: $\gamma = \cot\varphi + \tan(\varphi - \alpha) = \cos\alpha/[\sin\varphi\cos(\varphi - \alpha)]$.

$\cot 30° = 1.732$; $\tan(30° - 10°) = \tan 20° = 0.364$

$\gamma = 1.732 + 0.364 =$ **2.10**: **(B)**. (1.73 is $\cot\varphi$ alone.)

Exam tip: For a given α, γ is minimum when $\varphi = 45° + \alpha/2$, the same condition as Merchant's minimum-energy solution with zero friction.

Q4 **Answer: (D) 15°**

Concept: ASA tool signature order: back rake α_b, side rake α_s, end relief, side relief, end cutting edge angle C_e, side cutting edge angle C_s, nose radius.

So 10° back rake, 12° side rake, 8° end relief, 6° side relief, **15° end cutting edge angle**, 20° side cutting edge angle and 0.8 mm nose radius: **(D)**.

Exam tip: In ORS, the principal cutting edge angle $\varphi = 90° - C_s = 70°$ here, and the auxiliary cutting edge angle $\varphi_1 = C_e = 15°$.

Q5 **Answer: 1234.9 (accept 1228.7 to 1241.0)**

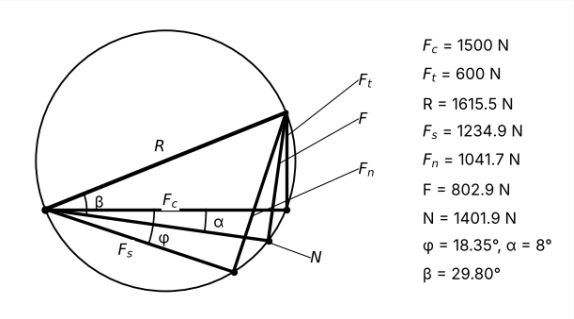
Concept: All the forces are components of the same resultant R, so in the Merchant's circle (see figure) $F_s = F_c\cos\varphi - F_t\sin\varphi$.

$\tan\varphi = 0.32\cos 8°/(1 - 0.32\sin 8°) = 0.3169/0.9555 = 0.3317$, so $\varphi = 18.35°$

$F_s = 1500\cos 18.35° - 600\sin 18.35° = 1423.7 - 188.9 =$ **1234.9 N**

Also $F_n = 1500\sin\varphi + 600\cos\varphi = 1041.7$ N, $F = 802.9$ N, $N = 1401.9$ N and $\mu = F/N = 0.573$.

Exam tip: F_s and F_n come from rotating (F_c, F_t) by φ; F and N come from rotating by α.



Q6 **Answer: (A) the shear angle decreases, the chip becomes thicker and the cutting force increases**

Concept: Merchant: $2\varphi + \beta - \alpha = 90°$, so $\varphi = 45° + \alpha/2 - \beta/2$.

A larger friction angle β gives a smaller φ. A smaller φ means a longer shear plane, a smaller chip thickness ratio r (thicker chip) and a larger shear area, so F_c rises: **(A)**.

Exam tip: Cutting fluids, polished rake faces and larger rake angles all raise φ and reduce power.

Q7 **Answer: 123.6 (accept 122.4 to 124.9)**

Concept: The velocity triangle closes: $V/\cos(\varphi - \alpha) = V_s/\cos\alpha = V_c/\sin\varphi$.

$\tan\varphi = 0.35\cos 5°/(1 - 0.35\sin 5°) = 0.3487/0.9695 = 0.3596$, so $\varphi = 19.78°$

$V_s = V\cos\alpha/\cos(\varphi - \alpha) = 120 \times 0.9962/\cos 14.78° = 119.54/0.9669 =$ **123.6 m/min**

Check: chip velocity $V_c = rV = 42$ m/min.

Exam tip: V_s can be larger than V; the shear power is $F_s V_s$ and the friction power is $F\,V_c$.

Q8 **Answer: 87.06 (accept 86.62 to 87.49)**

Concept: Taylor's equation $V\,T^n = C$.

$V_2 = V_1(T_1/T_2)^n = 100 \times (30/60)^{0.2} = 100 \times 0.8706$

$V_2 =$ **87.06 m/min**

Exam tip: Because n is small, doubling the tool life needs only about a 13 % cut in speed.

Q9 **Answer: (B) cubic boron nitride (CBN)**

Concept: The tool must keep its hardness at high temperature and must not react chemically with the work.

CBN is the second hardest material, is stable up to about 1400 °C and is chemically inert to iron, so it is used for hard turning of steels: **(B)**.

Diamond is harder but dissolves in iron (carbon diffusion) at high temperature, so PCD is used for non-ferrous metals and composites. HSS and uncoated carbides soften or wear fast on 60 HRC steel.

Exam tip: Order of hot hardness: HSS < carbides < ceramics < CBN < diamond.

Q10 **Answer: 104.3 (accept 103.3 to 105.3)**

Concept: For minimum cost, $T_{opt} = (1/n - 1)(t_c + C_t/C_m)$.

$T_{opt} = (4 - 1)(1 + 40/4) = 3 \times 11 = 33$ min

$V_{opt} = 250/33^{0.25} = 250/2.3968 =$ **104.3 m/min**

Exam tip: The speed for maximum production rate ($T = (1/n - 1)t_c = 3$ min) is always higher than the speed for minimum cost.

Q11 **Answer: (D) the cutting speed for a fixed tool life (often 60 min) on the material, as a percentage of that for a standard free-cutting steel**

Concept: Machinability is judged by tool life, forces/power and surface finish, but the common single-number rating uses tool life.

Machinability index = V_{60}(material)/V_{60}(standard) × 100 %, where the standard is AISI 1112 free-cutting steel (rated 100 %): **(D)**.

Exam tip: A material with index above 100 % (e.g. free-cutting brass, aluminium alloys) is easier to machine than the standard.

Q12 **Answer: 1.41 (accept 1.40 to 1.43)**

Concept: Cutting power = specific cutting energy × material removal rate (MRR = V f d).

$V = \pi DN = \pi \times 60 \times 500/60 = 1570.8$ mm/s

MRR = $1570.8 \times 0.2 \times 1.5 = 471.2$ mm³/s

P = $3 \times 471.2 = 1413.7$ W = **1.41 kW**

Exam tip: Specific energy (J/mm³) is numerically the same as the mean pressure $F_c/(f\,d)$ in N/mm².

Q13 **Answer: (B) carried away by the chip (typically about 70–80 %)**

Concept: Heat is generated in the primary (shear) zone, the secondary (tool–chip) zone and the tertiary (flank–work) zone.

At usual cutting speeds the chip moves away fast and carries most of the heat: **(B)**; roughly 10–20 % goes into the work and about 5–10 % into the tool.

Exam tip: The share taken by the chip increases with cutting speed, which is why high-speed machining can keep the work cool.

Q14 **Answer: 51.6 (accept 51.1 to 52.1)**

Concept: For constant T, $V f^{0.6}$ = constant, so $V_2/V_1 = (f_1/f_2)^{0.6}$.

$V_2/V_1 = 2^{0.6} = 1.5157$

Increase = **51.6 %**

Exam tip: Tool life is most sensitive to speed, less to feed and least to depth of cut, so to raise MRR without losing tool life, increase depth first, then feed.

Q15 **Answer: (A) both the chip thickness ratio and the shear angle increase, and the cutting force decreases**

Concept: From Merchant, $\varphi = 45° + \alpha/2 - \beta/2$, and $r = \sin\varphi/\cos(\varphi - \alpha)$.

A larger α gives a larger φ; a larger φ gives a shorter shear plane and a thinner chip, i.e. a larger r. The smaller shear area lowers F_c: **(A)**.

Exam tip: The limit to raising the rake angle is the strength of the tool wedge, which is why carbides and ceramics often use negative rake.

What next?

Day 59 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

Dig Career Thrust app

Telegram: daily PDFs

Next, Day 60/90: Week 10 Revision. Comment **GATE90** on our reel to get all links.
Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.